MLE4228 AY25/26 Semester 1
Lab 4 Report

Lu Jianyi A0336707Y

Lock De Ying Ethan A0271815A

Deng Yuntian A0336672X

Gao Zixuan A0336702J

1. INTRIDUCTION

Thermal actuators are devices that generate motion by converting thermal energy into mechanical deformation. They operate based on the expansion, contraction, or twisting of thermally responsive materials when subjected to temperature changes.Thermal actuators are components that produce linear movements through the expansion and contraction of thermally responsive materials.

Fishing line (Tensor stroke) actuator (Figure 1.1) is a type of thermal actuator. It's typically made from common nylon or polymer thread. When heated, the polymer chains inside the fishing lines try to relax and shorten along the direction of the twist. This causes the line to contract or rotate, generating motion. When it cools down, the material returns to its original length, completing one actuation cycle. When the actuator cools down, the polymer chains return to their original configuration, allowing the fiber to extend back to its initial length.

Fishing lines actuators became increasingly popular across various applications due to their unique capacity to contract or twist when heated, mimicking the behavior of artificial muscles. Offering different advantages in performance and cost, they are wide used in soft robotics, artificial muscles, and adaptive systems. (Figure 1.2)

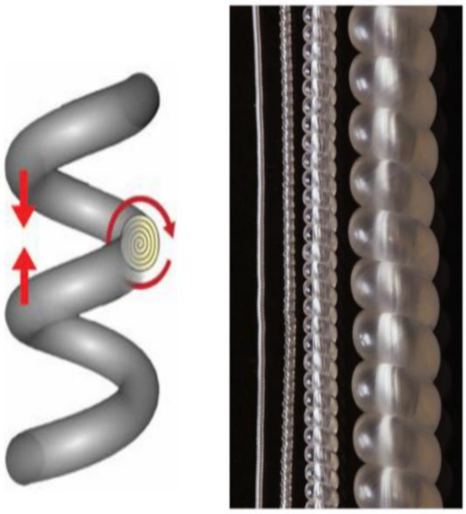

Figure 1.1 The schematic diagram of fishing line actuator

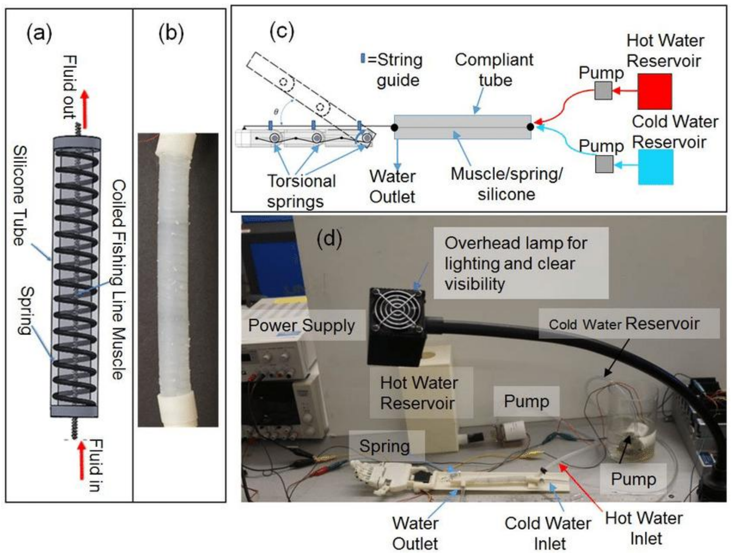

Figure 1.2 The application of fishing line (use hot water as heat source)

2. OBJECTIVE

This experiment aims at determining the optimal temperature for the suggested weight (170g) by applying a range of temperature on Nylon (D=0.27mm) and plot temperature versus actuation strain for a constant weight.

In addition, the experiment seeks to evaluate the maximum stress and strain of different actuator samples, including Nylon (0.27 mm), Polyethylene (PE, 0.27 mm), and Nylon (0.45 mm). By plotting actuation strain versus actuation stress of three material, the effects of material type and diameter on actuator performance can be analyzed using the stress–strain relationship.
A repeatability test for 0.27 mm nylon will also be conducted to assess the actuator's consistency, durability, and stability under repeated heating–cooling cycles. Furthermore, a multi-fiber actuation test will be performed using bundled 0.27 mm nylon fibers to investigate the effect of parallel configuration on overall actuation stress and strain, providing insights into the scalability and potential force enhancement of the actuator design.

3. MATERIALS AND METHODS

3.1 Materials and Equipment

Three types of fishing line (Figure 3.2) were used as thermal actuator material - Nylon (Both D=0.27mm and D=0.45mm) and Polyethylene (PE) (D=0.27mm). These materials are lightweight, flexible, and have excellent tensile strength and thermal stability. Nylon, in particular, is a semi-crystalline thermoplastic polymer — it consists of long molecular chains that can be stretched, twisted, or coiled to store mechanical energy.

Other tools for fabrication and various characterization tests include scissors, ruler, caliber, twister and temperature sensor, heat gun, weights, PC (running tracker software), smartphone (for video and image recording).

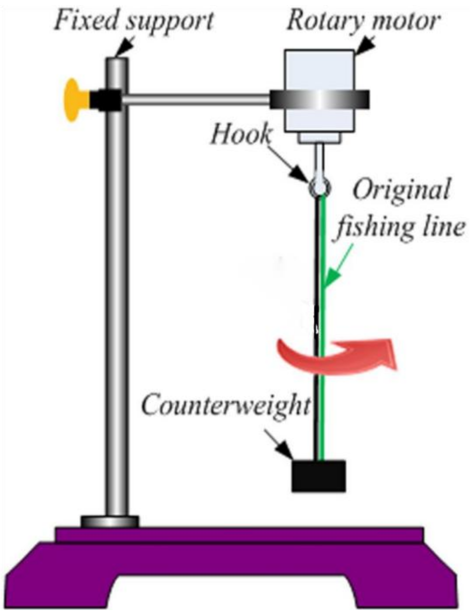

Figure 3.1.1 A set of experimental equipment Figure 3.1.2 Three kind of material

3.2 **Fabrication**

The design of a basic thermal actuator involves transforming a nylon fishing line into a twisted or coiled structure capable of contracting upon heating.

For helical fiber preparation, a 30 cm nylon fiber (D = 0.27 mm) was attached to a motorized twister on one end and a 20 g counterweight on the other to maintain tension. The fiber was twisted at a controlled speed until uniform coiling formed. Proper tension was crucial—excessive stress could cause breakage, while insufficient tension led to uneven or random curling. Continuous monitoring was required to stop the twisting promptly and avoid over-curling or deformation.

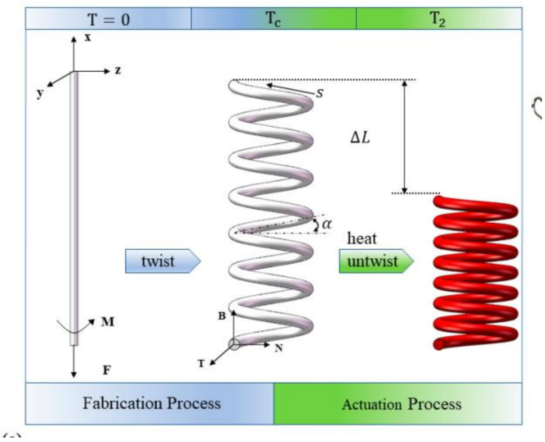

Figure 3.2.1 & Figure 3.2.2 Before and after twisting

3.3 **Experimental setup and method**

To determine the optimal temperature for the Nylon (D = 0.27 mm) actuator, a range of temperatures (50–200 °C) was applied under a constant load of 170 g. The heat gun was held 10–15 cm from the sample and moved evenly along the coil for 2–3 seconds per pass, followed by a 2 second pause. This heating–cooling cycle was repeated 3–4 times to ensure uniform heating without damaging the fiber. The maximum contraction (peak strain) at each temperature was recorded via video analysis.

For the second test, actuation stress and strain were measured for Nylon (0.27 mm), PE (0.27 mm), and Nylon (0.45 mm). Using the optimal temperature determined previously, different weights (170 g, 270 g, 370 g) were applied to obtain the corresponding stress–strain data. The same procedure was repeated for all materials to compare the effects of material type and diameter on actuation performance.

To evaluate repeatability, the most responsive configuration (Nylon 0.27 mm, 170 g) underwent at least ten heating–cooling cycles under identical conditions. The actuation strain and recovery behavior of each cycle were analyzed for consistency and potential degradation.

For the multi-fiber test, two 0.27 mm nylon fibers were bundled and tested using the same procedure to examine whether parallel configuration enhances actuation stress and strain compared to a single fiber.

4. <u>RESULT & DISCUSSION</u>

4.1 Temperature versus actuation strain for a constant weight

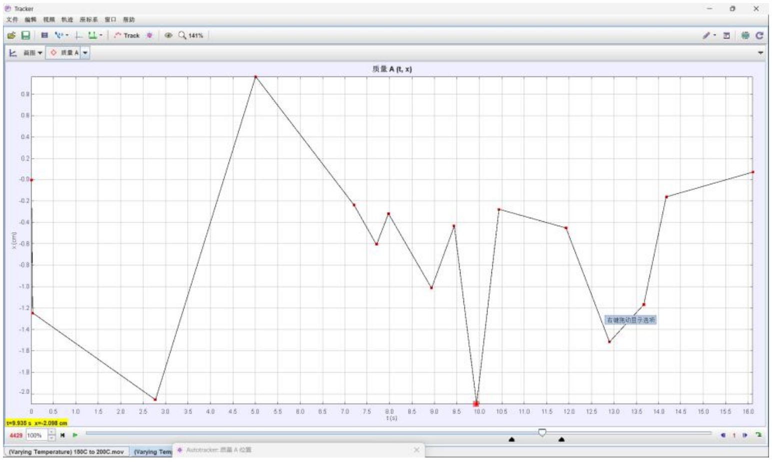

Figure 4.1.1 The displacement-time curve under 200°C (Max displacement= 2.098cm)

As shown in Figure 4.1.1, the *Tracker* software was used to analyze the video recorded during the first experiment, allowing us to obtain the displacement–time (x–t) graph of the end of the fishing line. In this graph, the x-axis represents the repeated experimental cycles, while the y-axis indicates the displacement of the fishing line's end. A negative displacement corresponds to an upward movement.

	50℃	100℃	150℃	200℃	250℃
max displacement (cm)	-0.35	-1.139	-1.516	-2.098	-1.197
original length (cm)	5.757	6.565	6.535	6.194	6.965
strain	-6.08%	-17.35%	-23.20%	-33.87%	-17.19%

Figure 4.1.2 Temperature - Strain form

To calculate the actuation strain, the maximum strain observed at each temperature was used, where L_0 = initial length (before actuation) and L = final length (after actuation).

$$Strain = \frac{L - L_0}{L_0}$$

As shown in the graph, the optimal actuation temperature for the Nylon filament (diameter = 0.27 mm) under a 170 g load is approximately 200 °C, producing a maximum actuation strain of 33.87%. The actuation strain increases with temperature up to 200 °C and decreases beyond this point.To present the results more clearly, we only include a representative x–t curve at 50 °C. The format of remaining x–t graphs (Both in this task and following tasks) are similar to this one.

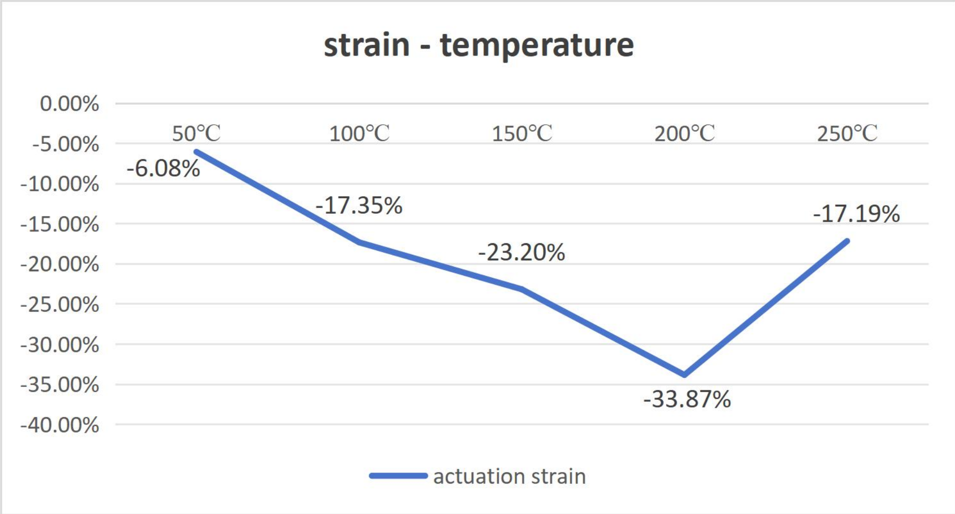

Figure 4.1.3 strain - temperature curve (50 °C - 250 °C)

DISCUSSION 4.1.1

Why the strain increased then decreased with increasing temperature?

Before reaching the optimal temperature, as temperature rises, internal polymer chains in nylon continue to relax and reorient, tightening the coil. This converts more radial expansion into axial contraction, so strain magnitude increases. The fiber behaves like a well-controlled muscle — higher temperature, stronger contraction. However, when the temperature exceeds the optimal point, excessive heating softens or partially damages the polymer. The material may undergoes plastic deformation, and internal stresses are permanently relaxed, causing the coil to loosen and reducing its ability to maintain strong contraction. Consequently, the strain decreases beyond the optimal temperature.

DISCUSSION 4.1.2

The possible error occurred when calculate strain and solution.

After analyzing all the displacement–time curves, we found that with increasing experimental cycles, most curves—especially those under higher stress—show a gradual upward shift. This indicates that the “original length” (equilibrium position) of the coil slowly increases over time. The likely reason is that the coil’s durability is limited, and repeated loading causes slight plastic deformation or creep, gradually stretching the coil. However, when calculating strain, the initial length at the beginning of the experiment was used each time, ignoring this small permanent elongation. As a result, the calculated strain values are slightly overestimated compared to the actual instantaneous deformation.

To reduce this error, more coil samples can be prepared, and each should be tested only once or twice to ensure the actuators remain in good condition. In addition, the reference (original) length can be updated continuously as the equilibrium position shifts upward during repeated loading, allowing the calculated strain to better reflect the actual deformation.

4.2 Find out actuation strain - stress relationship of three materials & rate the effect of diameter and material

An interesting behavior (Figure 4.2.2) was observed on Nylon (D=0.27m) under a 270 g load: the fiber first oscillated around its original length before gradually stabilizing at a slightly longer equilibrium position. Two steady states were identified—an initial plateau (State a) and a later extended plateau (State b)—both lasting for similar duration. For analysis, both the average and maximum strains of these states were plotted, where L_0 is the initial length, L the final length, and d the fiber diameter.

$$Strain = \frac{L - L_0}{L_0} \qquad Stress = \frac{F}{A} = \frac{mg}{\pi (d/2)^2}$$

	170g	270g(a)	270g (b)	370g
max displacement (cm)	-2.098	-0.669	-1.181	-1.121
original length (cm)	6.194	7.145	8.545	9.203
strain	-33.87%	-9.36%	-13.80%	-12.16%
average strain	-33.87%	-11.58%		-12.16%
stress (MPa)	29.1	46.2		63.4

Figure 4.2.1 Strain - Stress form for Nylon (D=0.27mm)

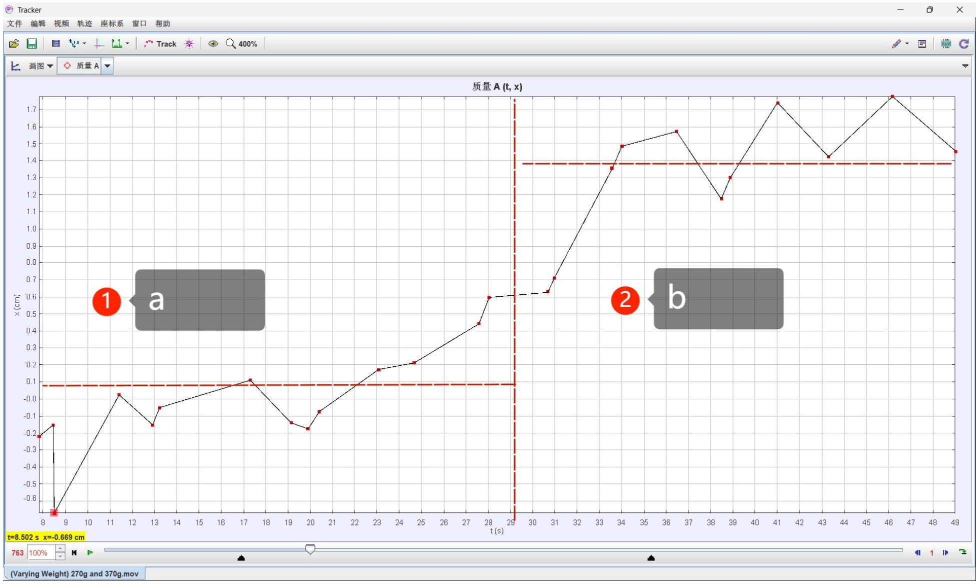

Figure 4.2.2 Two section when apply 270g on Nylon (D=0.27mm)

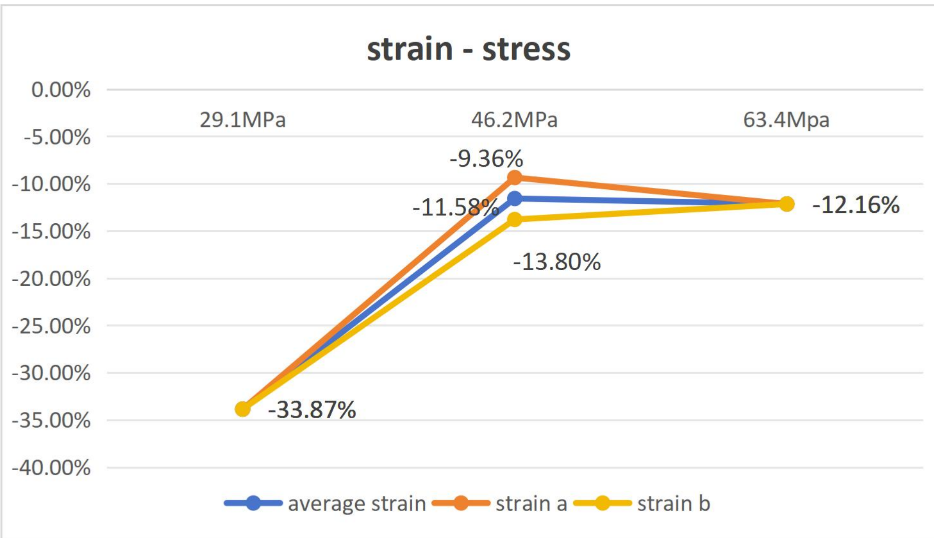

Figure 4.2.3 strain - stress curve of Nylon (D=0.27mm)

For the nylon sample, regardless of whether it is in state a or state b, the actuation strain reaches a relatively high value under a 170 g load. However, when the load is increased to 270 g and 370 g, the strain decreases sharply and then stabilizes at a low or moderate level. When the load is too low, the fiber may experience insufficient tension, which means the fiber doesn’t stay straight while twisting.It tends to fold or kink instead of forming a uniform helix. The maximum stress is 63.4MPa. Beyond that, the fiber fractured.

<u>DISCUSSION 4.2.1</u>

Why there are two states?

We think the reason is quite similar to DISCUSSION 4.1.2. but in this case, the coil deformation is more pronounced. Instead of a gradual increase, the coil quickly stretched to a new equilibrium position. This behavior suggests that under higher loading, the internal polymer structure of the nylon underwent partial plastic deformation or structural relaxation. As heating continued and the structure further relaxed, the coil extended and stabilized at a new equilibrium point, resulting in a second apparent maximum.

A. Effect of diameter

Same as the previous methods, we analyse the video of Nylon (D=0.45mm).

	170g	270g	370g
max displacement (cm)	-0.384	-0.432	-2.012
original length (cm)	6.972	6.882	8.476
strain	-5.50%	-6.27%	-23.73%
stress (MPa)	10.49	16.65	22.82

Figure 4.2.4 Strain - Stress form for Nylon (D=0.45mm)

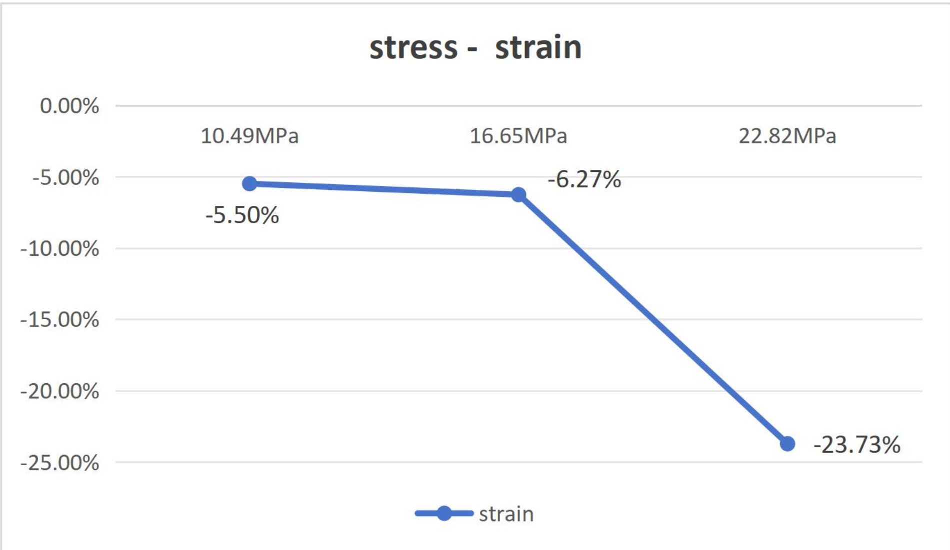

Figure 4.2.5 strain - stress curve of Nylon (D=0.45mm)

For the nylon fiber with a 0.45 mm diameter, it was observed that under relatively low applied stress, the actuation strain remained minimal and showed little variation with increasing load. However, when subjected to higher stress levels, the strain increased sharply, indicating a sudden enhancement in actuation response. Despite this, the fiber could not maintain stability under the high load and fractured shortly thereafter.

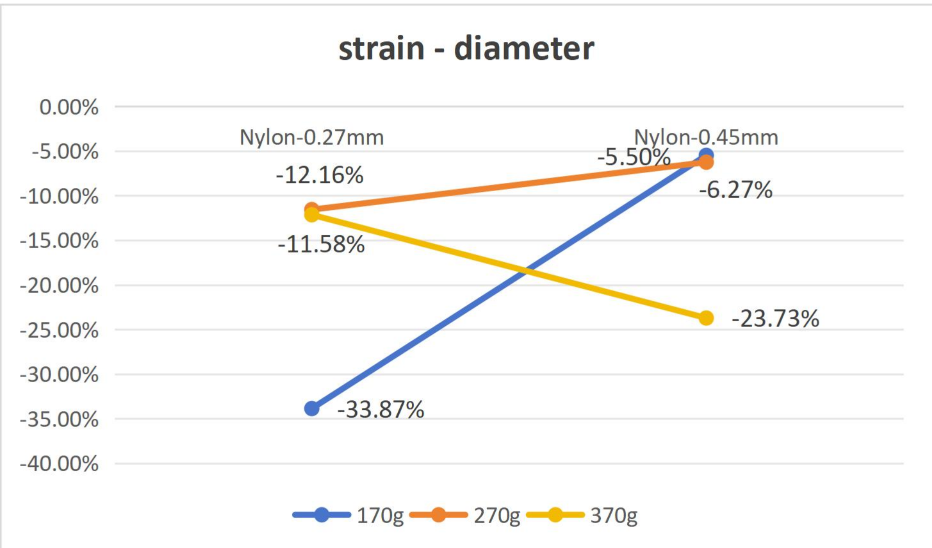

Figure 4.2.6 strain - diameter curve to show effect of diameter

To visualize the influence of fiber diameter on actuation performance, a diameter–strain graph was plotted, with each applied load represented as a separate data series. The results reveal that under low applied loads, the actuation strain decreases significantly with increasing fiber diameter. At intermediate loads (270 g), the difference in strain between the two diameters becomes smaller, though the thinner fiber still exhibits slightly higher actuation. Interestingly, when subjected to high loads, the trend appears to reverse — the thicker fiber exhibits a sudden increase in strain before failure.

DISCUSSION 4.2.2

For the 0.27mm fiber, why the strain drops sharply at first before stabilizing with increasing stress?

One possible reason is that the coils are loosely loaded at first, allowing the thermal expansion of the fiber to efficiently convert into axial contraction. When a higher load is applied, the tensile force from the mass exceeds the internal stress generated by the fiber's thermal contraction. As a result, the coil cannot contract fully, and the observed actuation strain decreases. Also, the higher stress may stretches and stiffens the coil, resisting contraction and limiting the actuator's ability to shorten. Once the coil reaches a new equilibrium between the applied mechanical load and thermal contraction, the strain stabilizes.

DISCUSSION 4.2.3

For the 0.45mm fiber, discuss the phenomena that the strain increase sharply with increasing stress, which seems abnormal.

For thicker fibers or multi-fiber bundles, the actuator is stiffer due to the larger cross-sectional area, so their coils resist deformation.Under low loads (170 g–270 g), the applied stress is insufficient to overcome this stiffness, so the coils is loosly loaded and barely engage, resulting in micro axial contraction and very small strain. When the load is increased to 370 g, the applied stress becomes large enough to activate the coil mechanics, allowing the radial expansion of the fibers to be efficiently converted into axial contraction, and producing noticeable actuation strain. This behavior indicates that thicker or stiffer actuators require a higher threshold load to achieve effective performance.

Integrate DISCUSSION 4.2.2 and DISCUSSION 4.2.3, thinner fibers are suitable for low-load applications, such as soft grippers for delicate objects or lightweight wearable actuators. Thicker fibers or multi-fiber bundles are better for high-load applications, such as lifting heavier loads in robotic arms, adjustable prosthetic devices, or tensioning mechanisms in adaptive textiles.

B. Effect of material

Same as the previous methods, we analyse the video of PE (D=0.27mm).

	170g
max displacement (cm)	-0.319
original length (cm)	6.729
strain	-4.74%
Stress (MPa)	29.1

Figure 4.2.7 Strain - Stress form for PE (D=0.27mm)

When a 170 g load was applied to the polyethylene (PE) fiber, it exhibited only a small actuation strain for several cycles before suddenly fracturing. To ensure this was not an experimental error, the test was repeated multiple times under identical conditions. However, the PE fiber consistently failed after a few actuation attempts, preventing further testing at higher loads.

The early breakage suggests that PE cannot tolerate high thermal or mechanical stress during actuation, likely due to its relatively low melting temperature and weaker molecular bonding compared with nylon. Consequently, only one valid data point was obtained, which represents the onset of failure rather than a stable actuation region. Therefore, a complete stress–strain curve for PE could not be established, and the material is deemed unsuitable for thermal actuation under the tested conditions. This behavior indicates that the PE material has poor thermal stability and low mechanical durability under repeated heating, making it unsuitable for high-temperature or cyclic actuation applications.

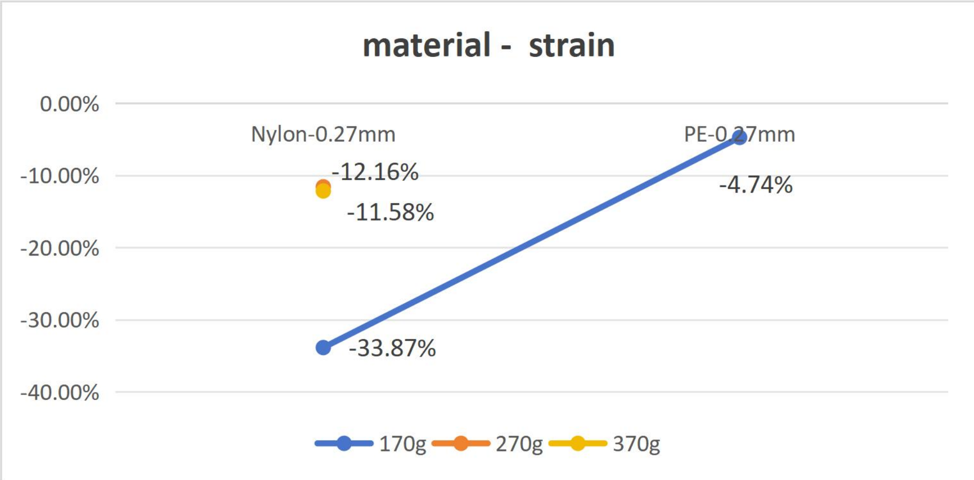

Figure 4.2.8 material - strain curve to show effect of material

Following the same approach, material type was plotted along the x-axis to compare actuation performance. For polyethylene (PE) and nylon, only the low-load case (170 g) is shown, as PE fibers consistently fractured before reaching 270 g or 370 g. The graph clearly demonstrates that nylon exhibits significantly higher actuation strain and greater durability than PE under the same low stress. At higher loads, PE cannot sustain the applied weight.

<u>DISCUSSION 4.2.4</u>

Discuss why this happened and the inspiration regarding the application of the two materials.

Nylon outperforms PE because it is tougher, more elastic, and can withstand higher stress without breaking, allowing larger actuation strain and repeated use. PE is weaker and fractures easily under moderate load, limiting its actuation. It may because their difference on polymer structure. One possible improvement is twisting them together. This suggests using nylon for durable, high-strain actuators (e.g., soft robotics or prosthetics) and PE for low-load, single-use, or lightweight applications.

4.3 Repeatability test of the most responsive group (Nylon D=0.27mm 200°C)

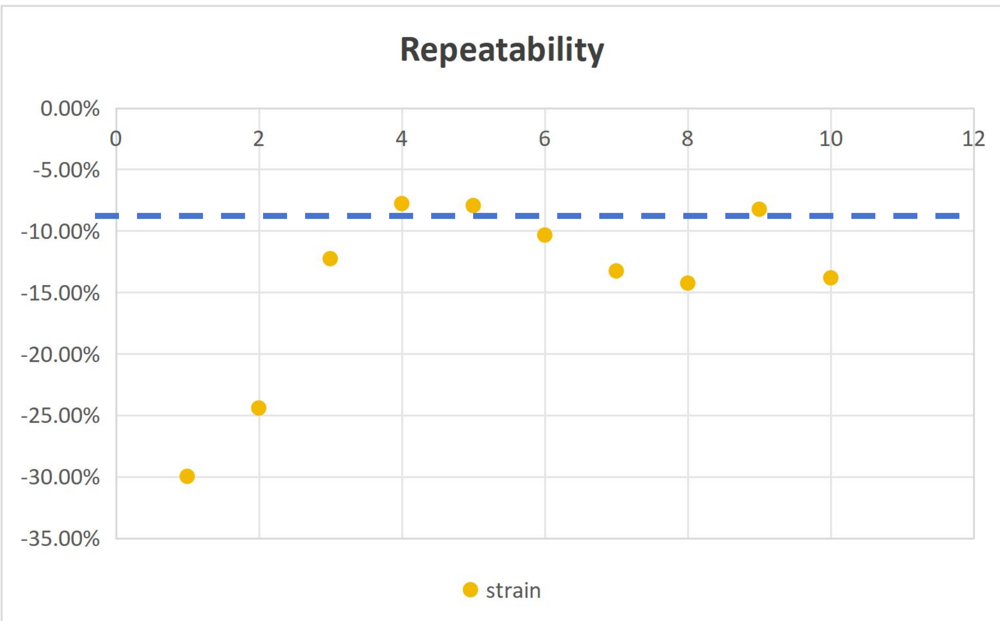

Figure 4.3.2 Actuation strain for 10 times on same condition

To evaluate the repeatability of the thermal actuator, we performed 10 consecutive heating–cooling cycles on the Nylon (D = 0.27 mm) sample under a constant load (170g). The actuation strain for each cycle, ε_i, was extracted from the recorded video data. The mean strain ($\overline{\varepsilon}$) and standard deviation (s) were calculated as follows:

$$\overline{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} (\varepsilon_i - \overline{\varepsilon})^2}$$

* Instead of $\frac{1}{n}$, we used $\frac{1}{n-1}$ here (known as Bessel's correction) to account for the fact that the sample mean is calculated from the same data set. This makes the calculated variance slightly smaller than the true population variance. Dividing by n−1 corrects this bias, providing a more accurate estimate of the population's true variability.

The repeatability was then quantified using the coefficient of variation (CV):

$$Repeatability\ (\%) = \frac{s}{\overline{\varepsilon}} \times 100$$

An interesting phenomenon was observed during the repeatability analysis. When all data points were included, the calculated repeatability was 56.00%. However, we noticed that the first two measurements were significantly higher than the rest. After removing these initial outliers and recalculating, the repeatability dropped to 30.29%, which is more reasonable. This suggests that the strain in the fishing line is initially unstable and higher but stabilizes over subsequent cycles, yielding higher repeatability once extreme early data are excluded. Additionally, the trend indicates that the actuator's performance decreases as the number of experimental cycles increases.

DISCUSSION 4.3.1
Why this happened?

One possible reason is that at the beginning, the fiber is in optimal condition, so the actuation strain is relatively high. As the number of cycles increases, minor material wear or fatigue may cause slight helical unwinding of the coil, leading to a reduction in strain. After this adjustment, the actuator reaches a new equilibrium, and the strain stabilizes at a steady level.

Another possible reason is based the mechanisms reported in Carter et al. (2014). During the first few cycles, the coiled nylon fibers experience residual stresses and structural settling, which cause the initial strain to be higher and less stable. As the actuator undergoes more cycles, the coils adjust and the polymer chains relax, resulting in more consistent and repeatable strain. Over many repeated cycles, however, minor creep or plastic deformation of the fibers can occur, gradually reducing the actuator's performance. This explains why the initial measurements appear extreme and why repeatability improves once these early cycles are excluded.

* Residual stresses : extra internal tension in polymer chains causes higher initial strain.
* Structural settling : minor coil adjustments during early cycles cause strain variation.

4.4 The result of the Multi-fiber Actuation test.

Due to differences in stress, fiber cross-sectional area, and performance, single- and multi-fiber data cannot be directly compared. Under low loads (170 g–270 g), the multi-fiber bundle showed negligible stretching, so higher loads (370 g–570 g) were considered. At 370 g, the strain remained very low; when the load increased to 470 g, the strain rose to a moderate level and remained relatively steady at 570g. When the load is too low, the fiber may experience insufficient tension, which means the fiber doesn't stay straight while twisting.It tends to fold or kink instead of forming a uniform helix. The maximum stress is 48.9MPa. Beyond that, the fiber fractured.

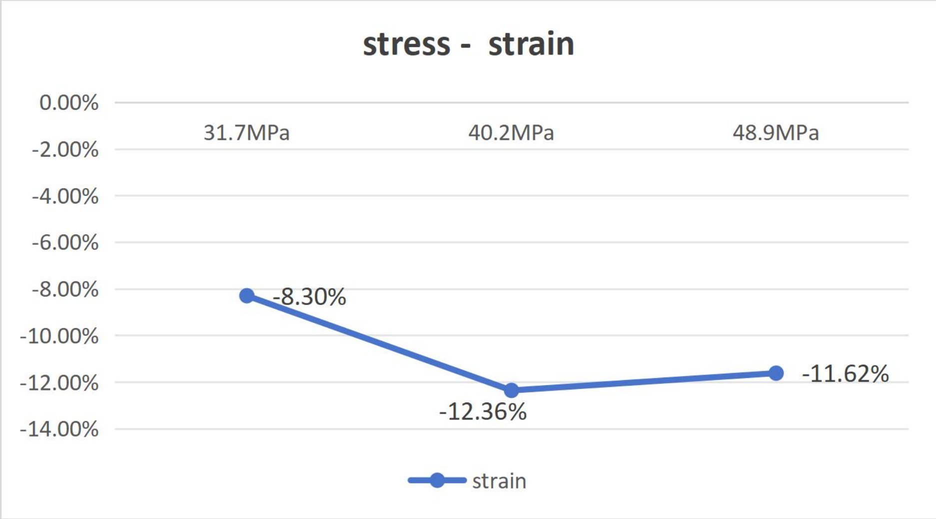

Figure 4.4.1 stress - strain curve for multi-fiber test

DISCUSSION 4.4.1

Discuss why multi-fiber's performance is poor under low stress, and discuss it's strength and drawbacks.

Based on Carter et al. (2014), the multi-fiber is stiffer then single one, which make it takes longer to heat up and cool down due to its larger volume and smaller surface area-to-volume ratio. Also, when the applied stress is too low (e.g., 170 g–270 g), the coils remain loose (because during fabrication, it's harder to insert enough twist per unit length in thicker fibers before buckling), so heating mainly causes radial expansion rather than effective axial contraction, leading to low strain. Compared to single fibers, multi-fiber actuators offer greater load capacity, stability, and repeatability due to distributed stress, but they also show lower strain and higher fabrication complexity from uneven heating and inter-fiber friction. Such multi-fiber configurations are suitable for applications requiring higher force output and stability, such as robotic joints or lifting mechanisms.

5. CONLUSION

In summary, the key findings of this experiment include that 200 °C is the optimal working temperature for the 0.27 mm nylon fishing line, the actuation strain of thinner fibers increases and then decreases with increasing stress, while the strain of thicker fibers increases and then reaches steady state. The results also show that material type significantly affects actuator performance. Regarding repeatability, the fishing line exhibited poor performance, especially during the initial cycles. Multi-fiber tests behaved similarly to the 0.45 mm nylon fiber tests, due to their greater thickness and stiffness.

These results indicate that the original objectives were fully achieved. Overall, the experiment was moderately successful, with clear trends observed despite some abnormal actuator behavior. For future work, improvements could include more precise measurements, optimized fabrication techniques, and tighter control of environmental variables.

6. REFERENCE

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5. Kim, S. H., Haines, C. S., Li, N., & Baughman, R. H. (2017). *Coiled polymer artificial muscles: A review of recent progress.* Advanced Materials, 29(11), 1603483.

APPENDIX

1. The methods used to analyse the video

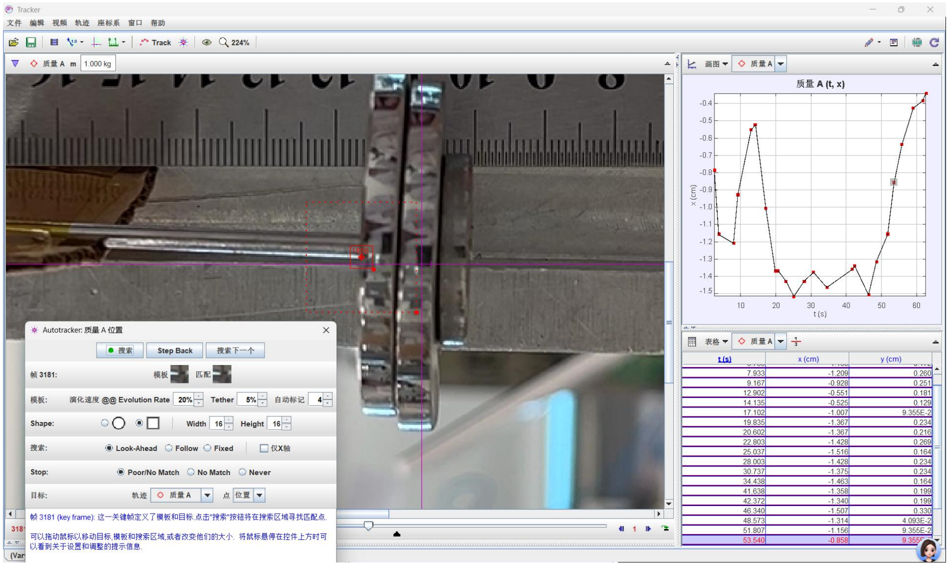

Figure 1.1 *Tracker* software used to analyse the video

Although the software has automatic track function, but it's easy for the algorithm to lose the track of the weights, especially when the heat gun approached the weights. So I tracked each frame manually, which was accurate but not detailed enough, because less data point were tracked.

2. The poor durability of PE

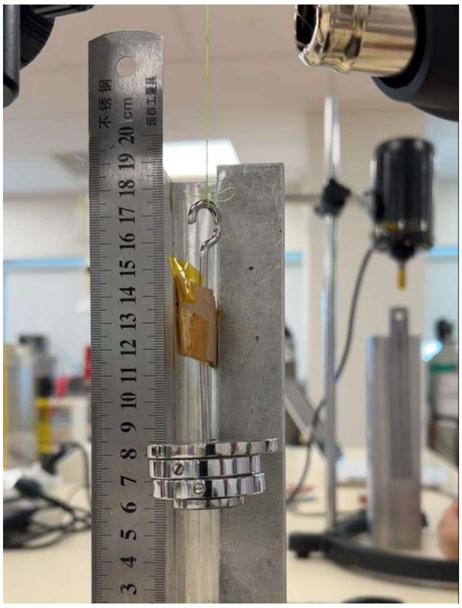

Figure 2.1 Few seconds before fracturing

As shown in the figure, the coil became uneven after multiple experimental cycles, indicating the occurrence of plastic deformation.

3. Reflection of the fabrication

The factors that may effect the quality of the fiber: rotation speed, exceed curling, uneven & fast heating.

a. The rotation speed determines how uniformly the fiber twists—too fast causes uneven tension or overwinding, while too slow may lead to loose or irregular coils.

b. Excessive curling or uneven, rapid heating can damage the polymer structure, leading to inconsistent coil geometry or partial melting.
c. Also, we noticed the orientation of the curling process varies. We assumed that the orientation of the spiral is mainly determined by how the fiber is stretched when knotted.

The fabrication process also included a pre-stretching step using a heat gun. We assumed it stretched the highly curled fiber to improve it’s performance and avoid over curling (?)

That may because during heating, the polymer first relaxes as internal stresses are released, and then shrinks as the molecular chains reorient and contract. This two-stage behavior plays a critical role in determining the final coil geometry and actuation capability of the fiber.

In practice, we met some challenges such as the metal wire on the spinning machine slipping off all the time, wasting a lot of time. This issue likely occurred because the tape failed to hold securely, possibly due to excessive load being applied during the coiling process.

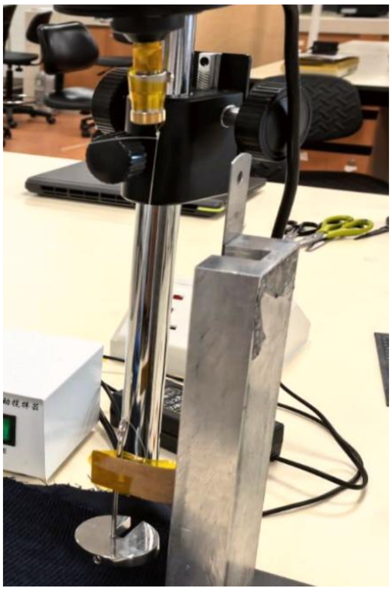

Figure 3.1 Before twisting

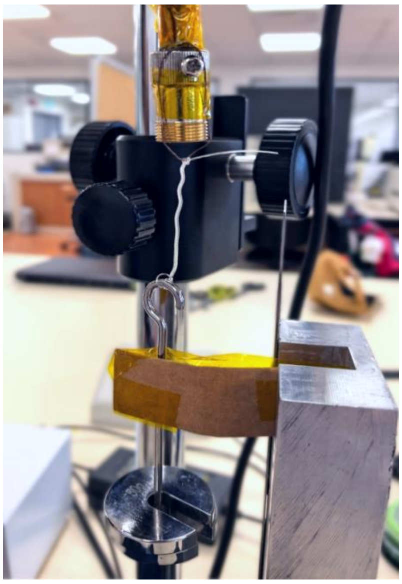

Figure 3.2 After twisting

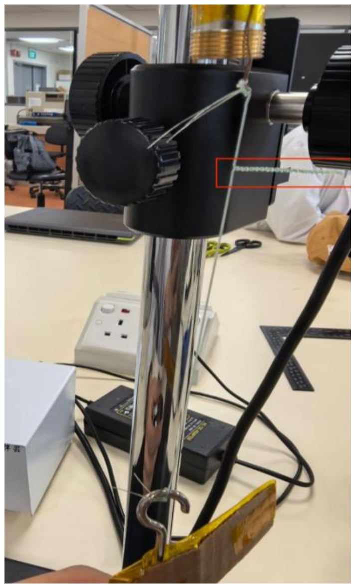

Figure 3.3 irregular spiral

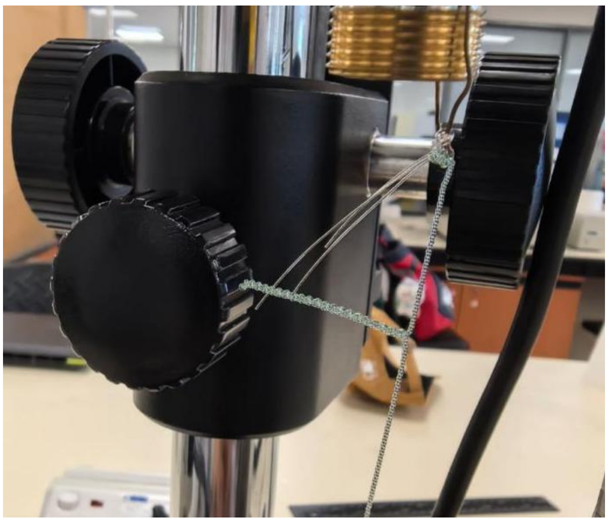

Figure 3.4 Too less load during fabrication caused problem